

Tuesday, 27/1/2026

Fun with Damped Harmonic Oscillation

1 A Rocking Jar

We will do a simple hands-on experiment to model and understand underdamped oscillations a bit better. To do so, we will use our smartphone and its sensors as a data-collecting device. You will need the **Physics Toolbox Sensor Suite** app which is available both on the [Apple App Store](#) and [Google's Playstore](#). Please download this app on at least one smartphone in your group.

The basic experiment we want to do is to understand the damped oscillations of a simple household jar. You can see a demo here to get a sense what you have to do:

<https://tinyurl.com/rockingjar>

I would like you to repeat the experiment and be bit more quantitative about it:

- a Make sure the app is downloaded and you should select the *Gyroscope* option. Play with it a little to understand how it works, what it is recording, and which axes of the phone correspond to x , y and z .
- b When you are ready to collect data, politely ask a TA to come provide you with a jar. You can collect the data under the supervision of the TA and then hand back the jar. Please do not take too much time since there are many groups and limited jars and limited time.
- c To collect data, insert the phone in the jar and then press the red **Start Recording** button to start recording your data. Let it stabilize. Now, give the can a little nudge with a small initial angular displacement so it starts oscillating. Wait long enough till the oscillations die out, and then stop the recording and transfer the data to your laptop/PC. The data will be in the form of a csv file containing measurements of angular velocities as a function of time.
- d Construct a simple mathematical model assuming the oscillator obeys underdamping with a damping force proportional to its velocity. Analyze the data (you can use any scientific computing resource and math techniques that you feel are appropriate. Excel or Google sheets is okay as well) and get numerical estimates of the oscillation frequency (what we called ω_u) and the linear damping coefficient (what we called γ in class). Further use this to infer the natural frequency of oscillation ω_0 .
- e Plot against your data the analytical solution (make sure you plot the appropriate quantity) of the underdamped oscillator to see how well this analysis went. Remember the underdamped oscillation model gave the following solution for the position,

$$x(t) = Ae^{-\gamma t/2} \cos(\omega_u t + \phi) . \quad (1)$$

- f You probably would notice the linear damping model is not a perfect model for explaining the dying out oscillations. It's good, but it's not *great*. Think why this is and what could be done to make this better. If you have a hypothesis and want to test it, ask the TA for the jar to record data again (for example, to compare a smooth vs rough surface).

Questions continue on the back...

2 Gotham Needs You!

Batman is looking to revamp his batmobile and one aspect he needs advice on is how to build a good shock-absorber system. In particular, you have been asked to find the required viscous damping coefficient in the system such that it absorbs shocks and returns it to the equilibrium position the fastest.

You model the shock-absorber as a one-dimensional massless spring which is connected to the batmobile's payload of mass m . Recalling from your PYL1002 class, you know the 1-D equation of motion of a body of mass m undergoing damped oscillations (with the resistive force proportional to its velocity) is given by

$$m \frac{d^2 x}{dt^2} + b \frac{dx}{dt} + kx = 0, \quad (2)$$

or

$$\frac{d^2 x}{dt^2} + \gamma \frac{dx}{dt} + \omega_0^2 x = 0, \quad (3)$$

where $\gamma = b/m$ is the damping coefficient and $\omega_0 = \sqrt{k/m}$ is the natural frequency of oscillation.

Hopefully you remember from class that a displaced mass returns and settles to the origin the *fastest in the case of critical damping*. Batman won't take your word for it, so you gotta prove it to him.

- (a) As a warm-up, write down the general solution $x(t)$ for the three possible cases of (i.) underdamping when $\gamma < 2\omega_0$, (ii.) over/heavy damping when $\gamma > 2\omega_0$ and (iii.) critical damping when $\gamma = 2\omega_0$. In each case, you will of course, have two undetermined constants which would depend on the initial conditions.
- (b) Let us focus on the overdamped/heavily-damped solution for this part. From your answer in part (a.), write the overdamped solution as,

$$x_{\text{overdamped}}(t) = C_1 \exp(-\mu_1 t) + C_2 \exp(-\mu_2 t), \quad (4)$$

with $\mu_1 > \mu_2 > 0$. In the above equation, C_1 and C_2 are unknown constants given by initial condition. In the analysis to follow, you can take both $C_1, C_2 \neq 0$. What are μ_1 and μ_2 in terms of γ and ω_0 ? **Which term in Eq. (4) (the one with μ_1 or μ_2 ?) is important for late time behavior as $x(t) \rightarrow 0$?**

- (c) Now for the case of Critical Damping which corresponds to $\gamma = 2\omega_0$. In class I conjectured that even $x = Cte^{-\gamma t/2}$ is also a solution for our differential equation. Verify this explicitly. This would make the most general solution of the form,

$$x_{\text{crit}}(t) = (C_1 + C_2 t) e^{-\gamma t/2}. \quad (5)$$

- (d) Now show that the critically damped oscillator returns and settles to the origin faster than the overdamped case (for the same initial conditions).
Hint: akin to what we did in part (b.) for overdamping, think about the behavior of the oscillator as $x(t) \rightarrow 0$ in the critical damped case.
- (e) Okay, now all we are left to do it to demonstrate is that the critical damped oscillator returns and *settles* to the origin even faster than the underdamped case too. To do this, again think about how the mass returns to the origin in the critical damped situation and compare with with how the amplitude of the lightly-damped case falls with time.
Hint: remember that exponential suppression envelope in the lightly-damped case?

3 Still Time?

Kudos on getting here. One more quick question. Now, suppose we have two pendulums (aka pendula), A and B , with identical geometry, but different masses, m_A and m_B . They are in air, so there is some (light) damping. It is observed that pendulum A damps to one-half its initial amplitude in 100 s, and pendulum B damps to one-half its initial amplitude in 1000 s. If $m_A = 9$ kg, what is m_B ?

That's all for now, folks! More next time.